

Joint Communication-Motion Planning for UAV Swarm against Jamming with Multi-Agent Deep Reinforcement Learning

Zhenxin Guo, Yiming Liu, Yipeng Wang, Yue Meng, Baoling Liu
State Key Laboratory of Networking and Switch Technology,
Beijing University of Posts and Telecommunications, Beijing, China
Email: {zhenxinguo, liuyiming, wangyipeng, mengyue, blliu}@bupt.edu.cn

Abstract—In this paper, we investigate the joint communication-motion planning problem for unmanned aerial vehicle (UAV) swarm in the presence of jammers. Specifically, we consider a cluster-based UAV swarm architecture, where multiple cluster member (CM) UAVs transmit messages to a cluster head (CH) UAV through air-to-air links affected by malicious jammers. Our objective is to maximize the sum uplink rate of the UAV swarm by optimizing the trajectories and the transmit power of all UAVs. To achieve this goal, we formulate a joint multi-UAV trajectories and transmit power optimization problem under speed, transmit power, trajectories and received signal-to-interference-plus-noise ratio (SINR) constraints. In order to solve the problem, we establish a Markov decision process (MDP). For the multi-agent environment and the high-dimensional continuous action space, we adopt a multi-agent twin delayed deep deterministic (MATD3) policy gradient-based algorithm. Simulation results show that the proposed scheme can effectively improve the sum uplink rate of the UAV swarm compared to the baseline schemes.

Index Terms—UAV swarm, joint communication-motion planning, anti-jamming, deep reinforcement learning.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs), with their mobility, flexibility and line-of-sight (LoS) links, can be used in the airspace networking infrastructure, providing a new approach to creating more flexible networks [1]. However, the open nature of air-to-air wireless channels makes the communication links between UAVs vulnerable to malicious jamming attacks [2], which can lead to degraded communication quality, link loss, even more disruption of the UAV swarm network.

To address above issue, several anti-jamming communication solutions have been developed [3]–[5]. In [3], the authors studied the frequency hopping technique, where communication nodes continuously change the frequency to avoid the malicious jamming attacks. The authors in [4] proposed a joint channel and power allocation strategy to address the malicious jamming attacks issues in UAV

networks. The authors in [5] proposed a channel selection strategy to resist the malicious jamming attacks. While all of the aforementioned methods focus on anti-jamming solutions through optimizing communication parameters, it is worth noting that the inherent mobility of UAVs enables them to mitigate jamming through the strategy of spatial avoidance [6].

The trajectory design of the UAV anti-jamming has been extensively explored. In [7], the authors investigated the UAV flight path planning and power allocation problem to maximize the throughput of the UAV system, where a UAV acts as a relay node to relay information to the ground under malicious jamming attacks. In [8], the authors designed the trajectory of a UAV to achieve the secure communication between a UAV and ground base stations under multiple dynamic eavesdroppers and jammers at unknown locations. Similarly, the authors in [9] designed an anti-jamming path planning strategy for multiple UAVs connecting with the ground base stations in order to minimize the flight time of the UAVs to complete the mission. Above works mainly focus on anti-jamming communication scenarios between UAVs and ground base stations. To address anti-jamming problem for inter-UAV communication, the authors in [10] considered a UAV-UAV pair communication under jammers and investigated the path planning problem to minimize the flight time. However, the problem of anti-jamming for UAV swarm communications remains less explored. The larger number of UAVs in a swarm makes the joint optimization of their trajectories and communication parameters more challenging, posing greater difficulties in mitigating jamming compared to a UAV-UAV pair communication.

Motivated by the above concerns, we investigate the anti-jamming problem of UAV swarm under multiple malicious jammers. The UAV swarm includes a cluster head (CH) UAV and multiple cluster member (CM) UAVs. In our scenario, UAV swarm is on a mission, flying from starting points to end points, while ground-based jammers interfere with their wireless links. Considering the CM UAVs need to collect data and forward them to CH UAV during the execution of the mission, we optimize the uplink transmission

This work was supported in part by Young Elite Scientists Sponsorship Program by CAST under Grant No. 2023QNRC001, in part by National Natural Science Foundation of China (No. 62001050 and No. 62341112), and in part by the Fundamental Research Funds for the Central Universities (Grant No. 2023RC95). (Corresponding author: Yiming Liu)

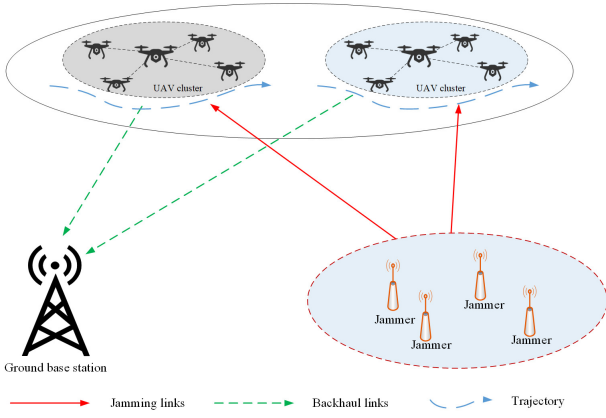


Fig. 1: UAV swarm under malicious jamming.

process. To maximize the sum uplink rate of UAV swarm under the malicious jamming attacks, we propose a joint communication-motion planning strategy, taking into account speed, transmit power, trajectories and received signal-to-interference-plus-noise ratio (SINR) constraints. Then we use the multi-agent twin delayed deep deterministic (MATD3) policy gradient-based algorithm to solve it. Simulation results show that the proposed scheme can effectively improve the sum uplink rate of the UAV swarm compared to the baseline schemes.

The rest of this paper is organized as follows. In Section II, we provide the system model and problem formulation. In Section III, a joint communication-motion planning algorithm based on MATD3 is proposed. Section IV discusses the simulation results. Finally, Section V summarizes this paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As depicted in Fig. 1, we investigate a clustered-based UAV communication system with multiple UAVs and jammers. In this scenario, the CM UAVs collect data and transmit messages to the CH UAV. Meanwhile, jammers on the ground intentionally emit jamming signals to disrupt the communication between UAVs. The set of UAVs is defined as $\mathcal{I} \triangleq \{0, 1, 2, \dots, i, \dots, I\}$, where 0 represents the CH UAV. The set of jammers is denoted by $\mathcal{J} \triangleq \{1, 2, \dots, j, \dots, J\}$. Let T denote the total mission completion time. Each UAV $i \in \mathcal{I}$ flies from a given start point $\mathbf{q}_i^{\text{start}} = [x_i^{\text{start}}, y_i^{\text{start}}, h_i^{\text{start}}]^T$ to an end point $\mathbf{q}_i^{\text{end}} = [x_i^{\text{end}}, y_i^{\text{end}}, h_i^{\text{end}}]^T$. Then, we discretize the duration T into $N = T/\Delta t$ time slots at a fixed time interval Δt . The set of time slots is defined as $\mathcal{N} \triangleq \{1, 2, \dots, n, \dots, N\}$. We assume that the positions of the UAVs are fixed at each time slot. Specifically, the position of the UAV $i \in \mathcal{I}$ at time-slot $n \in \mathcal{N}$ is denoted as $\mathbf{q}_i(n) = [x_i(n), y_i(n), h_i(n)]^T$. The position of the jammer $j \in \mathcal{J}$ on the ground is denoted as $\mathbf{q}_j = [x_j, y_j, h_j]^T$. Since UAVs can only get an estimate of the location of the jammer, the location information of the jammer is uncertain. We model the uncertain location of the jammer as a hemisphere,

where $\mathbf{q}_{j_c} = [x_{j_c}, y_{j_c}, h_{j_c}]^T$ represents the centre of the hemisphere. Hence, we can get

$$x_j = x_{j_c} + \Delta\sigma_{x_j}, y_j = y_{j_c} + \Delta\sigma_{y_j}, h_j = h_{j_c} + \Delta\sigma_{h_j}, \quad (1)$$

where $(\Delta\sigma_{x_j}, \Delta\sigma_{y_j}, \Delta\sigma_{h_j}) \in \Delta\sigma_q$ represents the deviation between the true position and the estimated position.

$$\Delta\sigma_q \triangleq \{(\Delta\sigma_{x_j})^2 + (\Delta\sigma_{y_j})^2 + (\Delta\sigma_{h_j})^2 \leq Q_j^2, \Delta\sigma_{h_j} \geq 0\}, \quad (2)$$

where Q_j represents the radius of the jammer j uncertainty hemisphere region.

A. Communication Model

Considering the free space propagation, the air-to-air communication is a LoS channel as in [11]. Therefore, the channel power gain between the CM UAV $i \in \mathcal{I}$, $i \neq 0$ and the CH UAV can be denoted by

$$h_{i,0}(n) = g_0(d_{i,0}(n))^{-\alpha} = \frac{g_0}{\|\mathbf{q}_i(n) - \mathbf{q}_0(n)\|^\alpha}, \quad (3)$$

where g_0 is the power gain with the reference distance of 1 meter [7], α is path loss index, and $d_{i,0}(n)$ is the distance between the CM UAV i and the CH UAV.

Communication channels from ground to air are primarily influenced by the large scale path loss. In the worst-case scenario, the strong LoS channels can be established between jammers and UAVs as in [11]. The channel power gain between the jammer $j \in \mathcal{J}$ and the UAV $i \in \mathcal{I}$ can be denoted by

$$h_{j,i}(n) = g_0(d_{j,i}(n))^{-\alpha} = \frac{g_0}{\|\mathbf{q}_j - \mathbf{q}_i(n)\|^\alpha}, \quad (4)$$

where $d_{j,i}(n)$ is the distance between the jammer $j \in \mathcal{J}$ and the UAV $i \in \mathcal{I}$. Hence, the aggregate interference received by the UAV $i \in \mathcal{I}$ from jammers can be given by

$$J_i(n) = \sum_{j=1}^J h_{j,i}(n)p_j, \quad (5)$$

where p_j is the transmit power of the jammer j .

The uplink SINR from the CM UAV $i \in \mathcal{I}$, $i \neq 0$ to the CH UAV and the downlink SINR from the CH UAV to the CM UAV $i \in \mathcal{I}$, $i \neq 0$ can be given by

$$\text{SINR}_{i,0}(n) = \frac{h_{i,0}(n)p_i(n)}{\sigma^2 + J_0(n)}, \quad (6)$$

$$\text{SINR}_{0,i}(n) = \frac{h_{i,0}(n)p_0(n)}{\sigma^2 + J_i(n)}, \quad (7)$$

where $p_i(n)$ is the transmit power of the UAV $i \in \mathcal{I}$, σ^2 is the power of additive white Gaussian noise (AWGN).

Then, we assume that the CM UAVs transmit messages to the CH UAV using orthogonal frequency division multiple access (OFDMA) technique, where the uplink backhaul bandwidth B is divided into M equal orthogonal subchannels with bandwidth $b = B/M$. M represents the number of CM UAVs. We consider that the jammers perform jamming of all frequencies of the UAV swarm. Accordingly,

the uplink data rate between the CM UAV $i \in \mathcal{I}$, $i \neq 0$ and the CH UAV at time-slot n can be expressed as

$$R_{i,0}(n) = b \log_2 \left[1 + \text{SINR}_{i,0}(n) \right], \quad (8)$$

B. Flying Model

In time slot n , the UAV $i \in \mathcal{I}$ is travelling at speed $v_i(n)$ in direction $\theta_i(n)$. Then, in the next time slot, the position of UAV i is given by

$$x_i(n+1) = x_i(n) + v_i(n) \Delta t \cos \theta_i(n), \quad (9)$$

$$y_i(n+1) = y_i(n) + v_i(n) \Delta t \sin \theta_i(n), \quad (10)$$

The maximum and minimum flight speed of UAVs are defined as v_{max} and v_{min} , respectively. The maximum distance a UAV can move within a fixed time interval Δt is d_{max} , which can be calculated as $d_{max} = v_{max} \times \Delta t$. Additionally, to prevent collisions between UAVs, it is important that the distance between UAVs is greater than the specified safety distance d_{min} . At the same time, UAV have to reach the destination within the mission time. Therefore, we derive the following constraints related to trajectory

$$\|\mathbf{q}_i(n) - \mathbf{q}_i(n-1)\|^2 \leq d_{max}^2, \forall i \in \mathcal{I}, \forall n \in \mathcal{N}, \quad (11)$$

$$\|\mathbf{q}_i(n) - \mathbf{q}_{i'}(n)\|^2 \geq d_{min}^2, \forall i, i' \in \mathcal{I}, i \neq i', \forall n \in \mathcal{N}, \quad (12)$$

$$\mathbf{q}_i(1) = \mathbf{q}_i^{start}, \forall i \in \mathcal{I}, \quad (13)$$

$$\mathbf{q}_i(N) = \mathbf{q}_i^{end}, \forall i \in \mathcal{I}. \quad (14)$$

C. Problem Formulation

The communication rate among UAVs is easily disrupted under jamming conditions. The CM UAVs need to collect data and forward them to CH UAV during the execution of the mission. Therefore, our goal focuses on increasing the sum uplink rate of UAV swarm. We denote $\mathbf{P} = \{p_i(n), \forall i \in \mathcal{I}, \forall n \in \mathcal{N}\}$ as the transmit power variables and $\mathbf{Q} = \{\mathbf{q}_i(n), \forall i \in \mathcal{I}, \forall n \in \mathcal{N}\}$ as the trajectory variables. The optimization problem is formulated as follows

$$\begin{aligned} \mathcal{P} : \max_{\{\mathbf{P}, \mathbf{Q}\}} & \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^I R_{i,0}(n) \\ \text{s.t.} \quad & C1 : 0 \leq p_i(n) \leq P_{max}, \\ & C2 : v_{min} \leq v_i(n) \leq v_{max}, \\ & C3 : \|\mathbf{q}_i(n) - \mathbf{q}_i(n-1)\|^2 \leq d_{max}^2, \\ & C4 : \|\mathbf{q}_i(n) - \mathbf{q}_{i'}(n)\|^2 \geq d_{min}^2, \\ & C5 : \mathbf{q}_i(1) = \mathbf{q}_i^{start}, \\ & C6 : \mathbf{q}_i(N) = \mathbf{q}_i^{end}, \\ & C7 : \text{SINR}_{i,0}(n) \geq \gamma_{th}, \\ & C8 : \text{SINR}_{0,i}(n) \geq \gamma_{th}. \end{aligned}$$

Where S_{th} denotes an SINR threshold. Constraint C1 denotes the transmit power constraints of UAVs. Constraints C2-C6 are the trajectory constraints of UAVs. Constraints C7-C8 ensure that the uplink SINR and downlink SINR are above the γ_{th} .

III. JOINT COMMUNICATION-MOTION PLANNING FOR ANTI-JAMMING IN UAV SWARM

In this section, we reformulate the optimization problem as a Markov decision process (MDP) and solve it using the MATD3 algorithm.

A. MDP Formulation for UAV Communication-Motion Planning

The joint communication-motion planning problem is formulated as an MDP. Commonly, an MDP is described by $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$, where $\mathcal{S}$ denotes state, $\mathcal{A}$ denotes action, $\mathcal{P}$ denotes transition probability, $\mathcal{R}$ denotes reward respectively. In an MDP containing K agents, the agent k at time slot n will get a current state $s_k(n)$ from the environment and then it will take an action $a_k(n) \in \mathcal{A}$ based on a certain policy. According to the transfer probability $\mathcal{P} = \{s_k(n+1) | s_k(n), a_1(n), \dots, a_K(n)\}$, it can transition to a new state $s_k(n+1)$. Finally, the agent can get a reward $r_k(n)$ to judge the action. The MDP formulation of our model is below:

1) **State**: The state of each UAV agent i at time slot n is defined as

$$s_i(n) = \{\mathbf{q}_i(n), D_i(n), v_i(n), p_i(n)\}, \quad (15)$$

where $\mathbf{q}_i(n)$ denotes the location information of the UAV agent i , $D_i(n) = \|\mathbf{q}_i(n) - \mathbf{q}_i^{end}\|$ denotes the distance information of the UAV agent i from the target point, $v_i(n)$ denotes the speed of the UAV agent i , and $p_i(n)$ denotes the transmit power of the UAV agent i .

2) **Action**: Since the variables of the optimization problem are the power and speed vectors, we set the actions of each UAV agent as the flight direction, the flight speed, and the transmit power. Thus, the action space of each UAV agent i at time slot n can be denoted as

$$a_i(n) = \{\theta_i(n), v_i(n), p_i(n)\}, \quad (16)$$

where $\theta_i(n) \in [0, \pi]$, $v_i(n) \in [v_{min}, v_{max}]$, and $p_i(n) \in [0, P_{max}]$.

3) **Reward**: In our work, the objective is to optimize the sum uplink rate of UAV swarm under trajectory and communication constraints. Thus, the reward of each UAV agent is divided into four parts. Let $r_{rate}(n)$ represents the reward of the sum uplink rate of UAV swarm. Considering the limited power of the UAV, we set a distance penalize. Let $r_{dis}^i(n)$ represents the distance penalize of the UAV i from the target point. Let $r_{traj}(n)$ represents the trajectory constraints penalize. Let $r_{sinr}(n)$ represents the SINR constraints penalize. $r_{rate}(n)$, $r_{dis}^i(n)$, $r_{traj}(n)$ and $r_{sinr}(n)$ are defined respectively as

$$r_{rate}(n) = \sum_{i=1}^I R_{i,0}(n), \quad (17)$$

$$r_{dis}^i(n) = \|\mathbf{q}_i(n) - \mathbf{q}_i^{end}\|, \quad (18)$$

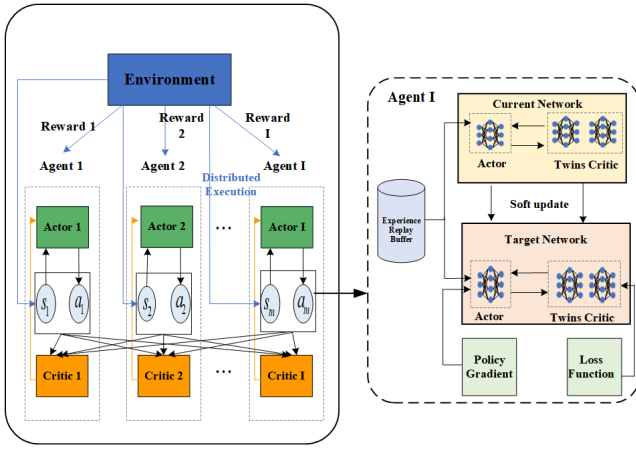


Fig. 2: The MATD3 framework of the multi-UAV.

$$r_{traj}(n) = \sum_{i=0, i \neq i'}^I \sum_{i'=0}^I \beta_{i,i'}(n), \quad (19)$$

$$r_{sinr}(n) = \sum_{i=1}^I \eta_i(n), \quad (20)$$

$$\beta_{i,i'}(n) = \begin{cases} 1 & \text{if } G_i(n) \geq d_{max}^2 \text{ or } F_{i,i'}(n) \leq d_{min}^2, \\ 0 & \text{otherwise.} \end{cases} \quad (21)$$

$$\eta_i(n) = \begin{cases} 1 & \text{if } SINR_{i,0}(n) \leq \gamma_{th} \text{ or } SINR_{0,i}(n) \leq \gamma_{th}, \\ 0 & \text{otherwise.} \end{cases} \quad (22)$$

where $G_i(n) = \|\mathbf{q}_i(n) - \mathbf{q}_i(n-1)\|^2$ and $F_{i,i'}(n) = \|\mathbf{q}_i(n) - \mathbf{q}_{i'}(n)\|^2$. Thus, the reward of each UAV agent i can be given by

$$r_i(n) = \lambda_1 r_{rate}(n) + \lambda_2 r_{dis}^i(n) + \lambda_3 r_{traj}(n) + \lambda_4 r_{sinr}(n). \quad (23)$$

where $\lambda_1, \lambda_2, \lambda_3$, and λ_4 denote the weight coefficients.

B. MATD3 Scheme

The MATD3 algorithm is a multi-agent algorithm. It solves the problem of environmental uncertain of single-agent in multi-agent scenario. Each UAV agent receives team rewards to encourage co-operation with each other.

Fig. 2 shows the framework of the MATD3. The agent has an action network $\mu(s|\theta^\mu)$ to generate actions and two critic networks $Q_1(s, a|\theta^{Q_1})$, $Q_2(s, a|\theta^{Q_2})$ to evaluate them. In addition, each agent also contains a target action network $\mu'(s|\theta^{\mu'})$, which has the same structure as the action network, and two target critic networks $Q'_1(s, a|\theta^{Q'_1})$, $Q'_2(s, a|\theta^{Q'_2})$, which are the same as the critic networks. The policy gradient of each UAV agent i follows the chain rule to update,

$$\begin{aligned} \nabla_{\theta_i^\mu} J(\mu_{\theta_i}) = & \mathbb{E}_{\mathbf{s}, \mathbf{a} \sim \mathcal{D}} [\nabla_{\theta_i^\mu} \mu_{\theta_i}(a_i | s_i) \\ & \times \nabla_{a_i} Q_i^\mu(\mathbf{s}, a_1, \dots, a_I) |_{a_i = \mu_{\theta_i}(s_i)}], \end{aligned} \quad (24)$$

where the experience replay buffer $\mathcal{D}$ is a tuple $(\mathbf{s}, \mathbf{s}', a_1, \dots, a_I, r_1, \dots, r_I)$, $\mathbf{s}$ and $\mathbf{s}'$ are the current state set

and the next state set of all UAV agents, respectively. θ_i^μ is the action network weights of the UAV agent i , μ_{θ_i} is the action network of the UAV agent i , a_i is the action of the UAV agent i , and s_i is the state of the UAV agent i . The critic network of each UAV agent i is updated by minimizing the loss function:

$$L(\theta_i^Q) = \mathbb{E}_{\mathbf{s}, a, r, \mathbf{s}'} [(y - Q_i^\mu(\mathbf{s}, a_1, \dots, a_I))^2], \quad (25)$$

$$y = r_i + \gamma \min_{u=1,2} Q_{i,u}^{\mu'}(\mathbf{s}', a'_1, \dots, a'_I) |_{a'_i = \mu'_{\theta_i}(s_i)}, \quad (26)$$

where θ_i^Q is the critic network of the UAV agent i , r_i is the reward of the UAV agent i , $Q_{i,1}^{\mu'}$ and $Q_{i,2}^{\mu'}$ are estimated target Q-value, and μ'_{θ_i} is the target action network of the UAV agent i .

The target action network and the target critic network of each UAV agent i are updated by means of soft updates, denoted as follows:

$$\theta_i^{\mu'} \leftarrow \rho \theta_i^{\mu} + (1 - \rho) \theta_i^{\mu'}, \quad (27)$$

$$\theta_i^{Q'_u} \leftarrow \rho \theta_i^{Q_u} + (1 - \rho) \theta_i^{Q'_u}, u = 1, 2. \quad (28)$$

where the soft coefficient $\rho \in (0, 1)$ defines how quickly the target networks update.

To achieve a balance between exploration and exploitation, we supplement the actor policy of each UAV agent i with noise collected from a noise process $\mathcal{U}$ to efficiently explore the action space:

$$a_i = \mu_{\theta_i}(s_i) + \mathcal{U}. \quad (29)$$

Based on the above preceding, **Algorithm 1** presents the MATD3-based joint communication-motion planning algorithm.

IV. SIMULATION RESULTS AND ANALYSIS

In this section, we illustrate the MATD3-based anti-jamming framework for UAV swarm through numerical simulations. First, the setting of simulation parameters is introduced. Then, the simulation results are analyzed.

A. Simulation setting

We consider a scenario involving 1 CH UAV, 3 CM UAVs, and 2 jammers. The flight area is a 3-D space of $600\text{m} \times 800\text{m} \times 250\text{m}$. The CH UAV and CM UAVs have starting positions of $(100,-300,150)$, $(50,-350,140)$, $(150,-350,140)$, $(50,-250,140)$ and end positions of $(500,400,150)$, $(480,400,140)$, $(400,330,140)$, $(400,400,140)$. The estimated location of the jammers are $(200,-50,0)$ with $Q_1 = 5\text{m}$ and $(350,50,0)$ with $Q_2 = 5\text{m}$. Table I lists the main simulation parameters.

Algorithm 1 MATD3-based communication-motion planning algorithm

- 1: Randomly initialize actor network μ and critic work $\mathbf{Q}$ with weights θ^μ and θ^Q .
- 2: Initialize target network μ' and $\mathbf{Q}'$ with weights $\theta^{\mu'} \leftarrow \theta^\mu$, $\theta^{Q'} \leftarrow \theta^Q$.
- 3: Initialize replay buffer $\mathcal{D}$.
- 4: **for** episode = 1,2,..., E **do**
- 5: Initialize random actions, and get observations $\mathcal{S}_{init}$.
- 6: **for** step = 1,2,..., N **do**
- 7: each agent i chooses $a_i = \mu_{\theta_i}(s_i) + \mathcal{U}$.
- 8: Agents execute actions $\mathcal{A}$, and get reward $\mathcal{R}$, new states $\mathcal{S}'$.
- 9: Store transition sample $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{S}')$ into $\mathcal{D}$.
- 10: **if** $\mathcal{D}$ is large enough **then**
- 11: Sample a random minibatch of B samples from $\mathcal{D}$;
- 12: Update actor network by the sampled policy gradient (24);
- 13: Update critic network by minimizing the loss (25);
- 14: Update the actor target network by (27);
- 15: Update the critic target network by (28).
- 16: **end if**
- 17: **end for**
- 18: **end for**
- 19: Output actor networks' weights for each agent θ_i^μ .

TABLE I: simulation parameters

Parameter	Default value	Parameter	Default value
T	100 s	λ_3	-1000
Δt	0.5 s	λ_4	-1000
d_{min}	50 m	γ_{th}	3 dB
g_0	-60 dB [11]	P_{max}	0.1 W
α	2	P_J	0.1 W
B	4 MHz	v_{max}	60 m/s
λ_1	1	v_{min}	10 m/s
λ_2	-0.01	σ^2	-109 dBm [11]

B. Simulation Results and Discussions

To evidence the performance of the MATD3 algorithm, we compare it with the MADDPG and DDPG algorithms. Fig. 3 compares the rewards obtained by MATD3, MADDPG and DDPG algorithms respectively. We can observe that the rewards increase as the number of training episodes increases. Also, we can see that the MATD3 algorithm obtains a better reward than the MADDPG and DDPG algorithms for the same learning rate. This is because the MATD3 uses two critic networks and adopts a centralized training strategy to evaluate the current state-action value more accurately.

Fig. 4 depicts the flight path of UAV swarm under jammers. Fig. 4(a) shows the 3-D flight trajectory of the UAV swarm. Fig. 4(b) shows the 2-D flight trajectory of the UAV swarm. We can observe the CH UAV flies as far away from the jammers as possible under the constraints of limited

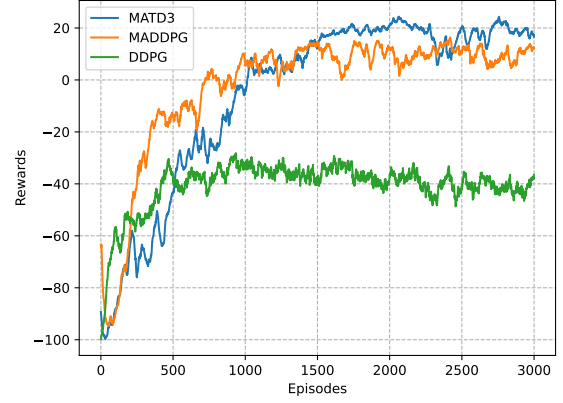
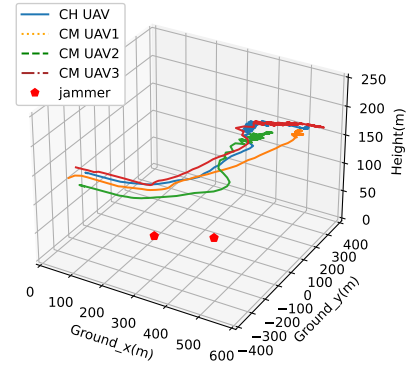
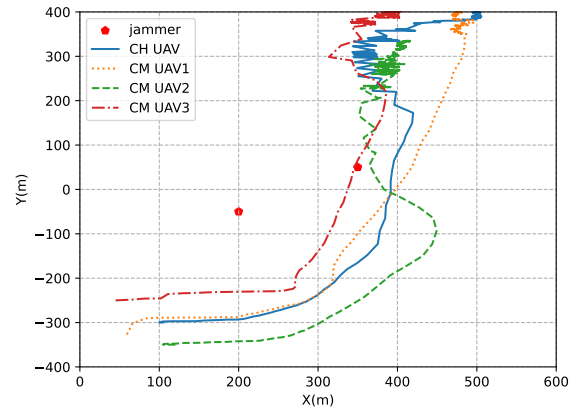


Fig. 3: Reward per episodes.



(a) 3-D flight trajectory



(b) 2-D flight trajectory

Fig. 4: UAV's flight trajectory under jammers.

mission time and destination point. Because our goal is to increase the sum uplink rate of the UAV swarm, the CH

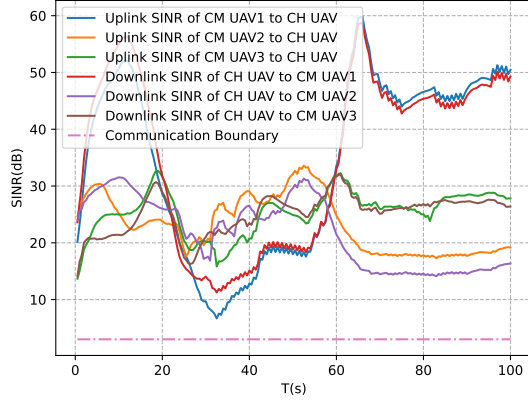


Fig. 5: The variation of SINR during the flight.

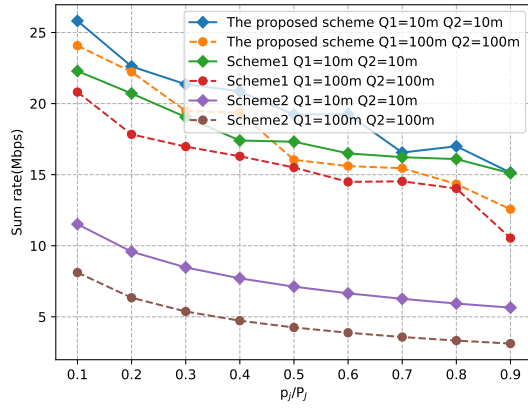


Fig. 6: The sum uplink rate of UAV swarm versus jamming power.

UAV deviates from the jammers as much as possible to mitigate the impact of jamming signal. In addition, due to the limitations of SINR and distance, the CM UAVs accompany the trajectory of the CH UAV as much as possible. Fig. 5 shows the variation of the uplink and downlink SINR between the UAVs during the flight. Both the uplink and downlink SINR are higher than γ_{th} .

To illustrate the performance of our proposed scheme, we compare it against two baseline strategies for evaluation, which are as follows:

- 1) Scheme 1: the flight path of the UAV swarm is optimized at a constant transmit power.
- 2) Scheme 2: the UAVs travels in a straight line from start to finish at a constant transmit power.

Fig. 6 depicts the sum uplink rate of UAV swarm changes as the jamming power p_j varies. The sum uplink rate of UAV swarm decreases with increasing jamming power. Meanwhile, we can find that our joint optimization scheme obtains higher the sum uplink rate of UAV swarm at the

same jamming power. Additionally, when the radius of the jammers uncertainty hemisphere region increases, the sum uplink rate of UAV swarm decreases. Therefore, incomplete jammer location information degrades system performance.

V. CONCLUSIONS AND FUTURE WORKS

In this paper, we proposed a joint communication-motion planning for UAV swarm anti-jamming. We investigated a joint multi-UAV trajectory and transmit power optimization problem under speed, transmit power, trajectories and received SINR constraints to maximize the sum uplink rate of the UAV swarm. Considering the problem is intractable using traditional approaches, we adopted an MATD3-based algorithm for obtaining feasible solutions. The simulation results demonstrate that our proposed anti-jamming scheme effectively enhanced the sum uplink rate of the UAV swarm compared to other baseline schemes. For future works, we will explore the joint sensing-communication-motion optimization to defend against smart jammers in the UAV swarm system.

REFERENCES

- [1] J. Li, X. Liu, G. Han, S. Cao and X. Wang, "TaskPOI Priority-Based Energy Balanced Multi-UAVs Cooperative Trajectory Planning Algorithm in 6G Networks," in *IEEE Transactions on Green Communications and Networking*, vol. 7, no. 2, pp. 1052-1065, June 2023, doi: 10.1109/TGCN.2022.3187097.
- [2] X. Sun, D. W. K. Ng, Z. Ding, Y. Xu and Z. Zhong, "Physical Layer Security in UAV Systems: Challenges and Opportunities," in *IEEE Wireless Communications*, vol. 26, no. 5, pp. 40-47, October 2019, doi: 10.1109/MWC.001.1900028.
- [3] K. Li, C. Wang, M. Lei, M. -M. Zhao and M. -J. Zhao, "A Local Reaction Anti-Jamming Scheme for UAV Swarms," *2020 IEEE 92nd Vehicular Technology Conference (VTC2020-Fall)*, Victoria, BC, Canada, 2020, pp. 1-6, doi: 10.1109/VTC2020-Fall49728.2020.9348489.
- [4] Z. Yin, Y. Lin, Y. Zhang, Y. Qian, F. Shu and J. Li, "Collaborative Multiagent Reinforcement Learning Aided Resource Allocation for UAV Anti-Jamming Communication," in *IEEE Internet of Things Journal*, vol. 9, no. 23, pp. 23995-24008, 1 Dec.1, 2022, doi: 10.1109/JIOT.2022.3188833.
- [5] F. Du, J. Li, Y. Lin, Z. Wang and Y. Qian, "Mean-Field Multi-Agent Reinforcement Learning for Adaptive Anti-Jamming Channel Selection in UAV Communications," *2022 14th International Conference on Wireless Communications and Signal Processing (WCSP)*, Nanjing, China, 2022, pp. 910-915, doi: 10.1109/WCSP55476.2022.10039304.
- [6] B. Duan, D. Yin, Y. Cong, H. Zhou, X. Xiang and L. Shen, "Anti-Jamming Path Planning for Unmanned Aerial Vehicles with Imperfect Jammer Information," *2018 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Kuala Lumpur, Malaysia, 2018, pp. 729-735, doi: 10.1109/ROBIO.2018.8665238.
- [7] Y. Wu, W. Yang, X. Guan and Q. Wu, "UAV-Enabled Relay Communication Under Malicious Jamming: Joint Trajectory and Transmit Power Optimization," in *IEEE Transactions on Vehicular Technology*, vol. 70, no. 8, pp. 8275-8279, Aug. 2021, doi: 10.1109/TVT.2021.3089158.
- [8] X. Zhang, S. Xu and L. Yang, "Learning to Navigate for Secure UAV Communication," *2021 IEEE Global Communications Conference (GLOBECOM)*, Madrid, Spain, 2021, pp. 1-6, doi: 10.1109/GLOBECOM46510.2021.9685830.
- [9] X. Wang, M. Cen Gursoy, T. Erpek and Y. E. Sagduyu, "Jamming-Resilient Path Planning for Multiple UAVs via Deep Reinforcement Learning," *2021 IEEE International Conference on Communications Workshops (ICC Workshops)*, Montreal, QC, Canada, 2021, pp. 1-6, doi: 10.1109/ICCWorkshops50388.2021.9473587.

- [10] Y. Wu, W. Yang and X. Guan, "UAV-UAV Communication Under Malicious Jamming: Trajectory Optimization with Turning Angle Constraint," *2020 International Conference on Wireless Communications and Signal Processing (WCSP)*, Nanjing, China, 2020, pp. 26-31, doi: 10.1109/WCSP49889.2020.9299835.
- [11] Y. Wu, X. Guan, W. Yang and Q. Wu, "UAV Swarm Communication Under Malicious Jamming: Joint Trajectory and Clustering Design," in *IEEE Wireless Communications Letters*, vol. 10, no. 10, pp. 2264-2268, Oct. 2021, doi: 10.1109/LWC.2021.3099128.
- [12] C. Zhang, S. Liang, C. He and K. Wang, "Multi-UAV Trajectory Design and Power Control Based on Deep Reinforcement Learning," in *Journal of Communications and Information Networks*, vol. 7, no. 2, pp. 192-201, June 2022, doi: 10.23919/JCIN.2022.9815202.
- [13] N. Gao, Z. Qin, X. Jing, Q. Ni and S. Jin, "Anti-Intelligent UAV Jamming Strategy via Deep Q-Networks," in *IEEE Transactions on Communications*, vol. 68, no. 1, pp. 569-581, Jan. 2020, doi: 10.1109/TCOMM.2019.2947918.
- [14] Y. Zhang, Z. Mou, F. Gao, J. Jiang, R. Ding and Z. Han, "UAV-Enabled Secure Communications by Multi-Agent Deep Reinforcement Learning," in *IEEE Transactions on Vehicular Technology*, vol. 69, no. 10, pp. 11599-11611, Oct. 2020, doi: 10.1109/TVT.2020.3014788.
- [15] Y. Zhang, Z. Zhuang, F. Gao, J. Wang and Z. Han, "Multi-Agent Deep Reinforcement Learning for Secure UAV Communications," *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, Seoul, Korea (South), 2020, pp. 1-5, doi: 10.1109/WCNC45663.2020.9120592.